



Apache Camel, a Popular Open Source Integration Framework

Apache Camel is a Java framework that simplifies the integration of complex systems. It saves developers from having to develop custom integration software for each system by offering a 'one application fits all' model.

Technology is growing at a rapid pace. Companies tend to move to newer, more stable, cost effective and scalable technologies. With this comes the challenge of integrating different systems that use different technologies, data formats and protocols. Integrating such complex systems from scratch is usually costly, tedious and seldom successful. Typically, it is more effective and less risky to make use of existing proven frameworks. Apache Camel is an open source integration framework, which focuses on integrating complex systems that use different protocols and technologies, seamlessly.

When should Apache Camel be used?

Enterprise application integration is necessary in every company. New technologies, formats and protocols make

it difficult to integrate systems. Writing custom code every time is not the solution as it often entails work for simple integrations. A simple integration like getting data from a Java API and sending it to an SFTP location on an event would work as shown in Figure 1.

As the complexities like sending/receiving data from many applications increase, where there are various protocols like SOAP, REST, SFTP and diverse systems involved, it becomes very difficult to manage and write the code. It also increases maintenance costs. Imagine a complex scenario where the data is retrieved from the database through the Java APIs and then sent to the REST component, which gives a response that is then sent to some SFTP location (as shown in Figure 2). This itself sounds very complex, but companies have even more complex scenarios than this as they want to integrate with all the applications.

Another scenario can be taking data from Twitter feeds and sending it to a system that does sentiment analysis on the data. The system then sends this analysed data to another application, which parses and sends the analytical charts to Slack Channel. Doesn't this seem rather complex?

It is pretty difficult to build such complex integration scenarios. The Apache Camel integration framework can be of help here. Camel just works like a jigsaw puzzle — we

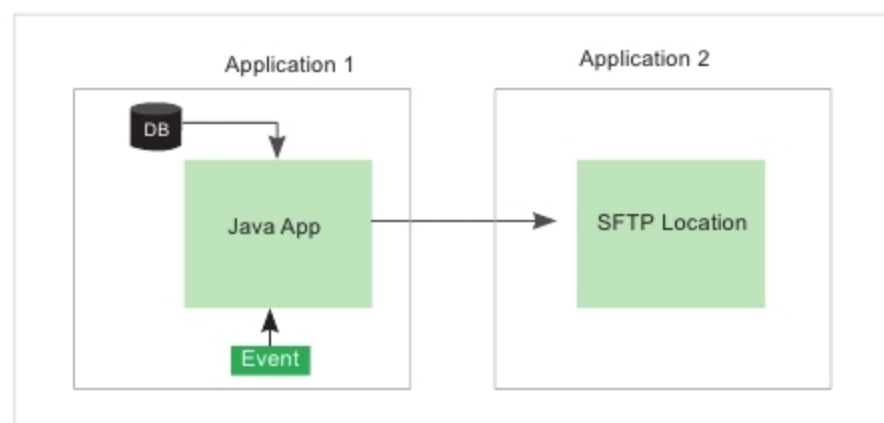


Figure 1: Simple integration